

The maximum number of stable matchings of order six is forty-eight

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Abstract

Let $f(n)$ denote the maximum number of stable matchings of a stable-marriage instance with n men and n women (Knuth 1976; Gusfield–Irving Open Problem #1). The known values were $f(1), \dots, f(5) = 1, 2, 3, 10, 16$; the value of $f(6)$ was open, with the lower bound 48 (a dihedral Latin instance) conjectured exact (OEIS A357271). We establish $f(6) = 48$. The proof is a reduction of the $\sim 10^{28}$ -instance search space to a finite space of *rotation schedules*: sequences of cyclic partner exchanges applied from the identity matching. Three elementary lemmas over the standard rotation theory show that every instance is dominated, in stable-matching count, by the *read-off instance* of its own schedule, and that a first-appearance labeling loses no schedules; the validity and bridge lemmas are machine-checked in Lean 4 (no `sorry`, standard axioms). The upper bound is then discharged by SAT certificates over a formula defined in Lean, asserting that a legal schedule reads off to at least 49 stable matchings: the canonical schedules are covered by 25,493 root cubes, refined by the solver’s splits into 321,492 cubes, whose 318,736 leaves were all refuted by CaDiCaL with every refutation checked by the formally verified LRAT checker `cake_lpr` (0 satisfiable; the journal is audited). The base formula, the cube lists, the certified cube set and every cube formula were re-derived from the Lean definitions and match what the solver saw. The Lean theorem that ties the formula to the count — every well-formed instance with at least 49 stable matchings satisfies some certified cube’s formula — is proved as well, so that $f(6) = 48$ is a single Lean theorem, `f6_eq_48_of_unsat`, whose only hypothesis is the unsatisfiability of the certified cube formulas (no `sorry`; axioms `propext`, `Classical.choice`, `Quot.sound`); the lower bound is a kernel-checked computation. An exhaustive enumeration that evaluates the exact stable-matching count of all 26,574,282,886 schedule nodes (maximum 48, roughly ten hours on a laptop) corroborates the result by a method sharing no code with the certificates; run at orders 3, 4, 5 it reproduces 3, 10, 16.

1 Introduction

A stable-marriage instance of order n equips n men and n women each with a strict preference order over the opposite side; a perfect matching is *stable* if no man and woman prefer each other to their assigned partners. Knuth [1] asked for the behaviour of $f(n)$, the maximum number of stable matchings over instances of order n ; Gusfield and Irving made it Open Problem #1 of their monograph [2]. Asymptotically $2.28^n \leq f(n) \leq 3.55^n$ [3, 4, 5]. Exact values have been scarce: $f(4) = 10$ is classical (Eilers’ uniqueness observation) and $f(5) = 16$ is due to Eilers [6]

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(obtained in 2022 by constraint programming and certified formally in the author’s companion work [7]), and

$$f(1), \dots, f(5) = 1, 2, 3, 10, 16.$$

For $n = 6$ the best lower bound has long been 48, attained by a Latin instance built from the dihedral group; OEIS A357271 records the conjecture that 48 is exact. The instance space of order six ($\sim 10^{28}$ after obvious symmetries) is far beyond direct search, and the best human-provable upper bound, $3.55^6 \approx 2005$, is forty times too large.

Result and method. We prove $f(6) = 48$. The key is a change of universe: instead of instances, we enumerate *schedules* — sequences of cyclic partner exchanges (“rotations”) applied from the identity matching, subject to the budget constraints that rotation theory imposes (at most 30 moved-man slots, at most 5 moves per man, no partner revisited). Each schedule *reads off* an instance whose preference lists are the participants’ partner trajectories, completed at the bottom. Three lemmas (Section 3) show: (i) every instance’s rotations, in elimination order, form a valid schedule; (ii) the read-off instance of that schedule has at least as many stable matchings (the *bridge lemma*); and (iii) a first-appearance labeling rule preserves at least one representative of every schedule without affecting read-off counts (a commutation normal form, used by the enumeration only, does likewise). The schedule space is small enough to certify: the canonical schedules are covered by 25,493 root cubes of a Lean-defined propositional formula asserting a read-off count of at least 49; refined by the solver’s splits into 321,492 cubes, all 318,736 leaves are refuted, each with a certificate checked by the verified checker `cake_lpr` (Section 5). With the dihedral lower bound, $f(6) = 48$ follows. An exhaustive enumeration that visits 26,574,282,886 schedule nodes and computes the exact stable-matching count of every node’s read-off instance, attaining maximum 48 (Section 6), corroborates the certificates by a method sharing no code with them.

Validation. Because the upper bound rests on computation, Section 5 records the audit of the certificate journal and the identity checks between the Lean-defined formula and cube lists and what the solver saw, and Section 8 documents the corroborating checks: the enumerator run at orders 3, 4, 5 reproduces $f = 3, 10, 16$ — the last value being formally certified (machine-checked in Lean, with SAT refutations verified by the formally verified checker `cake_lpr`) in the companion work [7], so a certified theorem serves as test oracle for the present method; an independently written enumerator agrees on maxima; disabling the symmetry reductions multiplies the tree by up to $48\times$ (a 2.66×10^9 -node check at order six) and never changes a maximum; and 1.3×10^9 local-search evaluations in two other search spaces also never exceeded 48. Section 9 describes the Lean 4 formalization of the reduction and states which links of the proof are Lean theorems.

2 Preliminaries

We use the structure theory of the stable-matching lattice and its rotations (Gusfield–Irving [2], Chapters 2–3) and record what we need.

Fact 1. *Call (m, w) a stable pair of an instance I if some stable matching contains it. For each man m , his stable partners ordered by his preference form his trajectory T_m : the man-optimal matching μ_0 gives each man the first entry of his trajectory, and the rotations of I , applied from μ_0 in any linear extension of the rotation poset, move each man strictly down*

his trajectory, one entry per rotation containing him, ending at the woman-optimal matching. Symmetrically each woman moves strictly up her trajectory. A rotation is a cyclic sequence $(m_0, w_0), \dots, (m_{k-1}, w_{k-1})$, $k \geq 2$, of pairs of the current matching, whose application gives m_i the partner w_{i+1} of m_{i+1} (indices mod k). The number of stable matchings equals the number of downsets of the rotation poset.

Fact 2. *Each man is moved by exactly $|T_m| - 1 \leq n - 1$ rotations, so the total move count satisfies $\sum_\rho |\rho| = \sum_m (|T_m| - 1) \leq n(n - 1)$; for $n = 6$: at most 30 slots, at most 5 moves per man, and every rotation moves at least two men.*

3 The schedule reduction

Definition 1 (Schedule). *Fix $n = 6$ and start from the identity matching. A schedule is a sequence of steps, each a cyclic move on $k \geq 2$ men (man m_i takes the current wife of m_{i+1}), subject to: total move count ≤ 30 , at most 5 moves per man, and no man ever moving to a wife he has already had, nor any woman receiving a husband she has already had.*

Definition 2 (Read-off instance). *Given a schedule S , the read-off instance $R(S)$ ranks, for each man, his trajectory in order at the top of his list, followed by the remaining women in a fixed canonical order; for each woman, her trajectory reversed at the top, followed by the remaining men canonically.*

Lemma 1 (Validity). *For any instance I of order 6, relabel women so that the man-optimal matching is the identity. Then the rotations of I , in any linear extension of the rotation poset, form a schedule $S(I)$.*

Proof. Rotations are cyclic moves on $k \geq 2$ men exposed in the current matching, and applying them in a linear extension realizes the lattice traversal, in which men move strictly down and women strictly up their trajectories (hence no repeats). The caps are Fact 2. $\square$

Lemma 2 (Bridge). *For every instance I of order 6 (women relabelled as above), $\text{sc}(I) \leq \text{sc}(R(S(I)))$.*

Proof. Write $J = R(S(I))$. The trajectories of $S(I)$ are the stable-partner sequences of I (Fact 1); thus for each man, J ranks his I -stable partners in the same relative order as I does, with all other women strictly below, and symmetrically for women. We claim every I -stable matching μ is J -stable, which proves the inequality. Every pair of μ is a stable pair, hence a trajectory pair on both sides. Let (m, w) , $w \neq \mu(m)$. If $w \notin T_m$: in J , m ranks w below all of T_m , in particular below $\mu(m)$; no block. If $w \in T_m$: then (m, w) is a stable pair, so $m \in T_w$; also $\mu(m) \in T_m$ and $\mu^{-1}(w) \in T_w$. Since J preserves relative I -order within trajectories on both sides, (m, w) blocks μ in J iff it blocks in I — and it does not. $\square$

Lemma 3 (Symmetry soundness). *(i) Exchanging two adjacent steps with disjoint man-sets leaves all trajectories, hence $R(S)$, unchanged. (ii) Relabelling men and women by a common permutation σ maps legal schedules to legal schedules and preserves the stable-matching count of the read-off: $\text{sc}(R(\sigma \cdot S)) = \text{sc}(R(S))$. Consequently an enumeration that (a) requires the new men of each step to be exactly the next unused labels as a set, and (b) rejects a step that*

is smaller, in a fixed total order on step shapes, than an earlier step separated from it only by man-disjoint steps,¹ still attains $\max_S \text{sc}(R(S))$.

Proof. (i) Steps with disjoint man-sets touch disjoint women (their women are their men’s current wives). (ii) Legality (budget, per-man cap, no revisit) is a property of the trajectories and is invariant under relabelling both sides by σ . The count equality is *not* an identity of instances: $R(\sigma \cdot S) \neq \sigma(R(S))$ in general, because the bottom completion of Definition 2 uses the fixed ascending order. Instead: in $R(S)$ the identity is man-optimal and the final matching woman-optimal, and every intermediate matching is stable (a pair (m, w) with m having held w earlier cannot block, since w would have to hold m again later), so each person’s stable partners in $R(S)$ are exactly their trajectory entries, in trajectory order. $R(\sigma \cdot S)$ ranks precisely these partners of $\sigma(R(S))$ on top in the same order, so the bottom-agnostic bridge of Section 9 (`count_le_of_orderPreserving`), instantiated at the relabelled instance $\sigma(R(S))$ and the relabelled schedule $\sigma \cdot S$, gives $\text{sc}(R(S)) = \text{sc}(\sigma(R(S))) \leq \text{sc}(R(\sigma \cdot S))$; applying σ^{-1} to $\sigma \cdot S$ gives the reverse inequality. This re-instantiation is how the Lean development proves the inequality it uses (at the chain schedule of an instance). For the consequence, relabel by first participation (any assignment within a step’s new set satisfies the set rule), and note rule (b) discards a sequence only in favour of a commutation-equivalent one with the same read-off instance. $\square$

4 A worked example

Order 3 is small enough to display the entire reduction. Consider the cyclic Latin instance I_3 (each row lists partners from most to least preferred):

m_0	w_0	w_1	w_2	w_0	m_2	m_1	m_0
m_1	w_2	w_0	w_1	w_1	m_1	m_0	m_2
m_2	w_1	w_2	w_0	w_2	m_0	m_2	m_1

It has $f(3) = 3$ stable matchings. In the man-optimal one every man gets his first choice: $\mu_0 = \{m_0w_0, m_1w_2, m_2w_1\}$; the woman-optimal one gives every woman hers. The rotation poset is a 2-chain of two rotations, each moving all three men, so the number of downsets is 3, as it must be.

Relabel women so that μ_0 becomes the identity ($w_0 \mapsto 0, w_2 \mapsto 1, w_1 \mapsto 2$). The two rotations become two applications of the same cyclic step $(m_0 m_2 m_1)$ — written (021) in min-first form — taking the matching through

$$(0, 1, 2) \longrightarrow (2, 0, 1) \longrightarrow (1, 2, 0).$$

Man m_0 ’s trajectory is (w_0, w_2, w_1) , which is exactly his (relabelled) preference order: at order 3 every pair of I_3 is a stable pair, so the trajectories exhaust the preference lists, the bottom completion is empty, and the read-off instance $R(S(I_3))$ is I_3 itself — the instance is a fixed point of the reduction, as every Latin extremal instance is.

The entire order-3 schedule space, enumerated under Definition 1 with the reductions of Lemma 3, has eight nodes:

¹Any total order on step shapes is sound. `gen_enum.c` uses the index order of its size-first shape table; the certificate campaign of Section 5 uses rule (a) only.

schedule	$sc(R(S))$	schedule	$sc(R(S))$
ε	1	(012)	2
(01)	2	(012)(012)	3
(01)(02)	3	(021)	2
(01)(12)	3	(021)(021)	3

The maximum is $3 = f(3)$, attained four times; $S(I_3) = (021)(021)$ is among the attainers, as Lemma 1 promises. Note that a maximizer need not use large steps — (01)(02), two transpositions, already attains it — a first glimpse of the all-size-2 phenomenon of Section 10.

5 The certificate campaign

The upper bound is discharged by SAT certificates over a formula defined in Lean. The formula Φ_{49} (`SchedCNF6.schedCNF6` 49 in `SchedCNF6.lean`; 84,882 variables, 2,709,212 clauses) describes a legal schedule of Definition 1 by a bounded sequence of frames (the matching and visit masks after each step), derives the read-off comparisons of Definition 2 from first visits, and asks for 49 pairwise distinct stable matchings of the read-off instance through selector slots in a fixed order. Satisfiability of Φ_{49} therefore says: some legal schedule reads off to at least 49 stable matchings.

Cubes. Φ_{49} is refuted cube by cube. A cube fixes a step prefix by unit clauses (`SchedCNF6.prefixUnits`); a *closed* cube additionally pins the schedule to that prefix (`Cubes6.stopUnits`). The root cubes are the 25,493 first-appearance canonical prefixes of length at most two (`Cubes6.canonicalCubes2`: the empty prefix closed, the 153 length-one prefixes closed, and 25,339 open length-two prefixes), which together cover every legal schedule satisfying rule (a) of Lemma 3; rule (b) is not used. A cube whose solve exceeded a certificate-size or time watchdog was split into its canonical one-step extensions (`Cubes6.extendCanon`) plus its closed copy: 1,804 roots were split at depth two and 952 of their children at depth three, giving 321,492 distinct cubes in all (by depth 0/1/2/3/4: 1 / 153 / 27,143 / 197,814 / 96,381), of which 2,910 are closed. The 318,736 leaves were each refuted by CaDiCaL 3.0.1 with LRAT output and each certificate checked by the formally verified checker `cake_lpr` [8]: 0 satisfiable, 318,736 **s VERIFIED UNSAT**. Solver time totalled 1,076,400 s (median 1.0 s per cube); the certificates, about 25 TB in all, were checked and deleted. The work ran on three machines with one solver build throughout: a Linux/x86-64 container (3,967 leaves and 235 splits, checker built from the published x86-64 assembly `cake_lpr.S`), an Apple M4 Max laptop (12,727 and 628) and an Apple M4 Pro Mac mini (302,042 and 1,893, checker built from the published arm64 assembly `cake_lpr_arm8.S`).

Journal and audit. Every run appends a record to a journal (cube id, depth, the SHA-256 of the DIMACS file the solver read, the SHA-256 and size of the LRAT file, solver and checker exit codes and output, timings). The audit (`cube_campaign.py --audit`) rebuilds the cube tree from the root generator and the recorded splits and checks that every leaf has a consistent verified record: 25,493 roots, 321,492 nodes, 318,736 verified, 0 missing, 0 bad. The journal is self-attested: a “verified” record is the driver’s transcription of `cake_lpr`’s verdict, and the certificates no longer exist; independent verification of a verdict means re-solving that cube, and the per-cube formula and certificate hashes let such a re-run be compared with the journal

record by record. The released journal (`f6/campaign/campaign.jsonl.gz`, 40,943,262 bytes) has SHA-256 `c9026b08045d8e8824c66d213bfa8eeb5336f118c22e030d42040136be7a8e4e`.

Identity with the Lean definitions. What the solver saw was re-derived from Lean after the campaign: the base formula printed by `export_sched_cnf` is byte-identical to the driver’s (SHA-256 `28421fb6...`); the 25,493 root cubes and the 295,999 split children printed by `export_cubes6` are byte-identical to the driver’s lists; the Lean-defined certified cube set `Cubes6.finalCubes` (the roots refined by the 1,804 and 952 recorded split ids, `SplitList6.lean`) equals the set of 318,736 verified ids; and all 321,492 journaled formula hashes recompute from the Lean-printed header, base body and unit clauses (`f6/lean_rehash.py`). As a positive control against vacuity, the formula with the dihedral schedule of Section 7 pinned is satisfiable at 48 slots and refuted at 49.

Theorem 1. $f(6) = 48$.

Proof. The dihedral Latin instance attains 48 (direct computation, three independent counters; OEIS A351413). For the upper bound let I be any instance. By Lemmas 1 and 2 (`validity_unconditional`) there is a legal schedule S with $\text{sc}(I) \leq \text{sc}(R(S))$, and by Lemma 3(ii) S may be taken to satisfy rule (a). Such an S fits one of the certified cubes, and if $\text{sc}(R(S)) \geq 49$ the schedule and its read-off would define a satisfying assignment of that cube’s formula. Every certified cube’s formula is unsatisfiable, so $\text{sc}(I) \leq 48$. $\square$

Remark 1 (What is a Lean theorem). *Every link in this proof is a Lean theorem except the checker verdicts. `validity_unconditional` (Section 9) is the reduction; the faithfulness theorem — every well-formed order-6 instance with at least 49 stable matchings yields a satisfying assignment of some certified cube’s formula, the analogue of `cube_faithful` in [7] — packages the schedule-level relabelling of Lemma 3(ii) (`Legal_relabel`, `firstApp_relabel`, `sc_le_readoffS_relabelSched`), the coverage of canonical schedules by `finalCubes` (`fits_final`), and the satisfaction of every clause family by the natural assignment (`cube_faithful6`); the assembly is `f6_eq_48_of_unsat` (Section 9). The identity checks above tie the certified formulas to the Lean definitions, and the certificates are checked by `cake_lpr`; the journal of verdicts is self-attested (Section 5).*

6 The enumeration (corroboration)

Independently of the certificates, an exhaustive enumeration evaluates every schedule node. The enumerator (`gen_enum.c`, ~200 lines of C) performs a depth-first search over schedules. The steps available at order 6 are the

$$\sum_{k=2}^6 \binom{6}{k} (k-1)! = 15 + 40 + 90 + 144 + 120 = 409$$

cyclic moves on 2..6 men in min-first form; state comprises the current matching, per-person trajectories and visit masks; the reductions of Lemma 3(a,b) and the caps of Definition 1 prune the tree. At *every* node the read-off instance is built (trajectory prefix + bottom completion) and its stable-matching count computed exactly by scanning all 720 perfect matchings. Sharding on the depth-1 step splits the work across cores. The same machine, parametrized by order, produces:

order	step shapes	nodes	$\max_S \text{sc}(R(S))$	$f(n)$
3	5	8	3	3
4	20	409	10	10
5	84	498,599	16	16
6	409	26,574,282,886	48	48

The full run (512 shards, 8 cores of an Apple M4 Max, ~ 10 hours) visits 26,574,282,886 nodes, evaluates all of them, and attains

$$\max_S \text{sc}(R(S)) = 48.$$

Because every node is evaluated, no monotonicity assumption about schedule extension is needed: for any instance I , the node $S(I)$ itself is evaluated (Lemma 3). This reproves the upper bound of Theorem 1 by a route that shares no code with the campaign of Section 5; its trust base is the C program and both halves of Lemma 3, neither of which is machine-checked, so we present it as corroboration of the certificates rather than as the proof.

7 The extremal instance

The unique-up-to-symmetry maximizer known at order 6 is Eilers’ Latin instance built from the dihedral group (OEIS A351413):

m_0	w_0	w_1	w_2	w_3	w_4	w_5	w_0	m_5	m_4	m_3	m_2	m_1	m_0
m_1	w_1	w_0	w_3	w_2	w_5	w_4	w_1	m_2	m_3	m_4	m_5	m_0	m_1
m_2	w_4	w_3	w_0	w_5	w_2	w_1	w_2	m_3	m_2	m_1	m_0	m_5	m_4
m_3	w_3	w_4	w_5	w_0	w_1	w_2	w_3	m_4	m_5	m_0	m_1	m_2	m_3
m_4	w_2	w_5	w_4	w_1	w_0	w_3	w_4	m_1	m_0	m_5	m_4	m_3	m_2
m_5	w_5	w_2	w_1	w_4	w_3	w_0	w_5	m_0	m_1	m_2	m_3	m_4	m_5

(rows list partners from most to least preferred; preferences are fully opposed: woman w ranks man m in position $5 - r$ whenever m ranks w in position r). Its structural profile is extreme in every measured respect: it has *fifteen* rotations, every one a transposition (size 2), for a total move count of 30 — the full budget $n(n - 1)$ of Fact 2 with every man moving the maximal five times, i.e. every man’s trajectory is his entire preference list. The rotation poset on these fifteen transpositions has 24 cover relations and exactly 48 downsets, which our three independent counters confirm as its stable-matching count. In the enumeration, only the shard family containing this schedule reaches 48; regions of the tree using any rotation of size ≥ 3 peak strictly lower.

8 Validation

Ground truth. The same enumerator, parametrized by order, reproduces every known value: order 3 (8 nodes, max 3), order 4 (409 nodes, max 10), order 5 (498,599 nodes, max 16). The last is decisive: $f(5) = 16$ is certified in the companion development [7] by a Lean 4 proof with SAT refutations checked end-to-end by the formally verified checker `cake_lpr` — a machine-checked theorem serving as test oracle for the present method.

Symmetry on/off. Disabling both reductions of Lemma 3 multiplies the tree ($4.5\times$ at order 4; $14\times$ at order 5; $48\times$ in a budget-16 order-6 check of 2,657,773,436 nodes) and never changes a maximum.

Independent implementation. A separately written Python enumerator (different pruning, different code path) agrees on the maxima at the budgets where it is feasible; a third, 40-line reference implementation written directly from Definition 1 and Lemma 3 reproduces the order-3 table of Section 4 node for node. A fourth route bypasses enumeration entirely: a SAT reformulation of the schedule space (bounded frames, first-visit-derived read-off comparisons, selector slots) refutes “some order-5 schedule has $\text{sc}(R(S)) \geq 17$ ” in four minutes, with the refutation checked end-to-end by `drat-trim` and `cake_lpr` — both an independent certificate-backed confirmation at order 5 and the pilot for the order-6 campaign of Section 5 (design notes in `REPLAY_DESIGN.md` in the repository).

Other search spaces. Before the enumeration, 9.7×10^7 hill-climbing evaluations in raw instance space and 1.2×10^9 evaluations in swap-schedule space (72 million valid schedules) never exceeded 48; an earlier maximal-only pass of the full enumeration (1.40×10^{10} exact counts) also attained 48.

Structural corroboration. All 450 sampled order-5 maximizers share a single rotation-poset skeleton; the dihedral order-6 instance uses the full budget of fifteen size-2 rotations; and across the enumeration only the shard containing the dihedral family reaches 48, with large-rotation shards peaking strictly lower — consistent with extremal instances being full-budget, all-size-2 phenomena.

9 Formalization in Lean 4

Lemmas 1 and 2 and the faithfulness theorem are formalized in Lean 4 with Mathlib in the accompanying development, atop the order-5 infrastructure of [7]. The reduction layer is eleven files from `SixBridge.lean` through `SchedCNF6.lean` (the last defines the campaign formula Φ_{49} as a Lean value): 5,230 lines and 243 theorems. The faithfulness layer is nineteen files (4,761 lines, 333 theorems) plus a 2,808-line data file, `SplitList6.lean`, holding the 2,756 recorded split decisions that define the certified cube set `finalCubes`. Its end statement, `f6_eq_48_of_unsat`, reads: if every cube of `finalCubes` has an unsatisfiable formula `cubeFormula 49 c`, then every well-formed order-6 instance has at most 48 stable matchings and the dihedral instance has exactly 48 (`dihedral6_count`, a kernel computation over the 720 permutations). Its proof runs: from $\text{sc}(I) \geq 49$ to the chain schedule of the woman-relabelled instance (`validity_unconditional`’s ingredients), relabelled by first participation (`Legal_relabel`, `firstApp_relabel`; the count is preserved by re-instantiating the bridge at the relabelled instance, `sc.le_readoffsS_relabelSched`, since R itself is not relabel-equivariant), to a canonical legal schedule with read-off count at least 49 (`exists_canonical_schedule`); such a schedule fits a cube of `finalCubes` (`fits_final`); and its natural assignment satisfies that cube’s formula (`cube_faithful6`: twelve clause-family lemmas and two unit-clause lemmas), contradicting the hypothesis. The end statement is unconditional: for every well-formed order-6 instance I there is a legal schedule S with $\text{sc}(I) \leq \text{sc}(R(S))$ (`validity_unconditional`), obtained by composing the validity chain with a *bottom-agnostic* form of the bridge (`count_le_of_orderPreserving`:

J dominates I whenever J ranks each person’s I -stable partners on top in I -order, with no condition on the rest of the lists) and a women-only relabelling that normalizes the man-optimal matching to the identity (`stableCount6.wrelabel6`, `manOpt.wrelabel6`). Lemma 3(ii) is formalized at the schedule level as described above; its commutation half (i) is validated computationally only (canonicalization on/off agreement) and is used by the enumeration alone. The certificate campaign of Section 5 needs the relabelling half (ii) — its root cubes cover only first-appearance canonical schedules — and not the commutation half (i). Every theorem is proved with no `sorry`, and `#print axioms` reports only the standard `propext`, `Classical.choice`, `Quot.sound`; continuous integration (on pushes to `main`, pull requests and manual dispatch) rebuilds the development from source with the standard Mathlib olean cache and re-checks the no-`sorry` and axiom conditions.

Design. Instances are pairs of 6×6 rank tables over `List Nat`; a matching is a list, stability is a decidable predicate, and the set of stable matchings `sms6 I` is computed by filtering the (structurally recursive, hence kernel-computable) permutation enumeration. The formalization uses *no* Gale–Shapley algorithm and *no* rotation machinery: rotations, which carry the classical proofs, never appear. This is a deliberate trade — the classical route (rotations exist, form a poset, covers eliminate single rotations) is textbook material but formalization-heavy, whereas everything the reduction actually needs can be rebuilt from elementary list lemmas.

The bridge lemma. The read-off instance is defined by a promotion operator: `readoff I` moves each person’s stable partners (a decidable predicate) to the top of their list, preserving relative order. The Lean statement of Lemma 2 is a direct case analysis: a would-be blocking pair of a stable matching in the read-off instance either involves a non-trajectory partner (ranked below everything relevant by promotion) or maps back to a blocking pair of the original instance.

The chain. Lemma 1 needs the lattice traversal. Instead of rotation elimination, the development constructs the maximal chain directly:

```
def chainFrom (I : Inst6) : Nat → List Nat → List (List Nat)
| 0, cur => [cur]
| fuel+1, cur => match (sms6 I).filter (strictDomB I cur) with
| [] => [cur]
| c :: cs => cur :: chainFrom I fuel (pickMin I c cs)
```

where `pickMin` selects, among the strict dominators of the current matching, one of *minimum total rank-sum*. This `argmin` choice does the work of rotation theory for free: any stable matching strictly between the current one and its successor would have a strictly smaller rank-sum than the successor, so every step is automatically a *cover* of the dominance order; the rank-sum strictly increases and is bounded by 30, so fuel 31 provably suffices and the chain ends at the woman-optimal matching. Completeness — every stable pair of I appears on the chain — follows from a *no-skip* lemma (on a cover $\mu \leq \nu$, no man passes over a stable partner) by squeezing each stable pair’s rank between the chain’s endpoints. The endpoint matchings themselves are obtained lattice-style: the meet of two stable matchings is stable (Knuth’s lemma, proved by blocking-pair extraction over lists), a fold of meets gives the man-optimal matching, and the woman-optimal side comes free by a transposition functor that swaps the two sides and inverts matchings.

Trajectories, budgets, cycles. Each clause of Definition 1 and Lemma 1 is an individual theorem about the chain:

clause	Lean theorem(s)
runs from man-optimal to woman-optimal	<code>chainFrom_head</code> , <code>chainFrom_last</code>
all elements stable matchings	<code>chainFrom_mem</code> , <code>chain_mem_perm</code>
consecutive elements are covers	<code>chainFrom_cover</code>
every stable pair realized	<code>chain_complete</code>
men descend their lists, exactly the stable partners	<code>traj_mem_iff</code> , <code>traj_rank_pairwise</code>
from optimal to pessimal partner	<code>traj_head</code> , <code>traj_getLast</code>
no revisits, either side	<code>traj_nodup</code> , <code>wtraj_nodup</code>
≤ 5 moves per man, ≤ 30 total	<code>traj_length_le</code> , <code>total_moves_le_30</code>
steps are disjoint cyclic swaps, size ≥ 2	<code>prevOwner_moved/ne/inj/getD</code>

The last row deserves a remark: the classical statement “every cover eliminates a single rotation” is *not* formalized, because it is not needed. For coverage of the schedule space it suffices that each step’s moved set decomposes into disjoint cycles of size ≥ 2 (the map “previous owner of my new wife” is a fixed-point-free injective self-map of the moved set — four short permutation lemmas); a multi-cycle step splits into consecutive single-cycle steps with identical trajectories.

Trust base. With the Lean-side identity checks of Section 5, what remains outside the kernel is, item for item, the trust base of [7]: the kernel and its three axioms; the forty-line definitions of instances, stability and the count at the top of `SixBridge.lean`; one LRAT checker; the two printers (`ExportSchedCnf.lean`, `ExportCubes6.lean`) together with the file identity; and the campaign journal, which is self-attested (a `verified` record is the driver’s transcription of `cake_lpr`’s verdict, and the certificates were deleted after checking), so that independently verifying a verdict means re-solving that cube from the Lean-printed formula. Neither the solvers nor any Python or C program is trusted.

10 Related work and outlook

The computational lineage on $f(n)$ is Eilers’ ($f(5)$, the uniqueness observation at $f(4)$, and the extremal-profile counts, which we replicated independently in [7]), building on Thurber [3] and Knuth [1]; the asymptotic bounds are [4, 5]. Certified computational values of comparable flavour include $R(4, 5) = 25$, the empty hexagon number, Keller’s conjecture and Schur number five; our companion paper [7] follows that tradition for $f(5)$, and Sections 9 and 5 carry it to the order-6 bound: the reduction and the faithfulness theorem are Lean theorems and the certificates are checked by `cake_lpr`, so that $f(6) = 48$ is, like $f(5) = 16$, one Lean theorem with checked certificates as its only hypothesis (Remark 1). The next open value, $f(7)$ (budget 42), had published lower bound 81 (Ong, Ang, Ho, Eilers, Marks and Buzi [9], improving Thurber’s 71 [3]); the accompanying repository raises it to 85 (`f7/1b85.txt`: two all-transposition schedules of length 20 found by local search). Certifying $f(7)$ appears to need either a much faster certified evaluator or new structural cuts: the schedule tree grows by three to four orders of magnitude.

An open problem. The extremal data of Sections 4 and 7 share a pattern. At orders 3 through 6 the maximum is attained by an *all-transposition* schedule (every step of size 2): (01)(02) at order 3; six transpositions (the full budget 12) at order 4; eight transpositions (16

of the budget 20) at order 5; fifteen transpositions (the full budget 30) at order 6. At the *even* orders the extremal schedule exhausts the budget — every man’s trajectory is his entire list — while at the odd orders 3 and 5 it does not: at $n = 3$ no legal full-budget all-transposition schedule exists, and at $n = 5$ every legal full-budget all-transposition schedule reads off to at most $11 < 16$ (by enumeration under Definitions 1–2). Whether the extremal schedule is sub-budget at every odd order is part of the observation, not a theorem. This is consistent with Eilers’ observation that the natural lower-bound constructions look exact only at even orders.

Conjecture 1. *For every $n \geq 3$, $f(n)$ is attained by the read-off of an all-size-2 schedule; for even n , by a full-budget one $(n(n - 1)/2)$ transposition steps).*

If true, computing $f(n)$ reduces to an optimization over sequences of transpositions — sorting-network-like objects — a space of roughly $(n(n - 1)/2)!^{1/2}$ rather than $n!^{2n}$, and structural insight into it could sharpen the 2.28^n – 3.55^n asymptotic band.

Artifact. All code, logs and verification instructions accompany the repository in its 2026-09-08 state (`STATUS.md` is the evidence ledger, `VERIFYING.md` the recipe): the campaign journal (SHA-256 in Section 5) with its audit, the Lean exporters and the re-hash script that reproduce the identity checks, and the enumeration, reproducible in ~ 10 laptop-hours. Re-solving the certificate campaign costs 1,076,400 solver-seconds (about 300 core-hours).

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